# IZT Race Technology

# System Architecture Development Specification

## Version 1.0 Draft

# 1. Purpose

Dokumen ini mendefinisikan arsitektur sistem untuk platform IZT Race Technology yang terdiri dari:

- Timing Hardware
- Timing Decoder Software
- Timing API
- Race Management System
- Participant Management System
- Live Leaderboard
- Result Calculation Engine

Tujuan utama sistem adalah menyediakan platform race timing yang scalable, real-time, audit-friendly, dan dapat digunakan untuk berbagai jenis event olahraga.

# 2. System Design Principles

## Principle #1 - Raw Data Must Never Be Lost

Data hasil pembacaan RFID merupakan sumber data utama.

Seluruh pembacaan EPC wajib disimpan sebagai raw timing data.

Raw timing tidak boleh dimodifikasi oleh operator.

## Principle #2 - Hardware Does Not Know Participants

Timing hardware maupun decoder software tidak menyimpan data peserta.

Timing system hanya mengetahui:

- EPC
- Timestamp
- RSSI

- Reader ID
- Device ID

Data peserta hanya berada pada platform web.

## Principle #3 - Race Results Are Derived Data

Hasil lomba bukan sumber data utama.

Hasil lomba dihitung berdasarkan:

- Raw Timing Data
- Participant Mapping
- Event Configuration
- Race Rules

Result dapat dihitung ulang kapan saja.

## Principle #4 - Multi Event Support

Satu platform harus dapat menjalankan banyak event secara bersamaan.

Contoh:

- Kostrad Run 2026
- Bandung Marathon 2026
- BRI Fun Run 2026

tanpa saling mempengaruhi data.

# 3. High Level Architecture

```
┌──────────────────────┐
│ RFID Hardware        │
└──────────┬───────────┘
           │
           ▼
┌──────────────────────┐
│ Decoder Software     │
│ (Windows / Linux)    │
└──────────┬───────────┘
           │ HTTP POST JSON
           ▼
┌──────────────────────┐
│ Timing API           │
└──────────┬───────────┘
           │
```

```
┌──────────┼──────────┐
▼          ▼          ▼

Raw      Participant   Result
Timing   Database      Engine

└──────────┬──────────┘
           ▼

 Live Leaderboard
           │
           ▼
      React Frontend
```

# 4. Component Responsibilities

## RFID Hardware

Responsibilities:

- Read RFID tags
- Generate EPC
- Generate RSSI
- Send reads to decoder

Must NOT:

- Store participant data
- Calculate ranking
- Calculate race result

## Decoder Software

Responsibilities:

- Receive EPC reads
- Buffer data
- Retry failed transmissions
- Send timing data to API

Must NOT:

- Store participant profiles
- Calculate race result
- Calculate leaderboard

## Timing API

Responsibilities:

- Receive timing data
- Validate requests
- Store raw timing
- Trigger result processing

Must be stateless.

## Race Management System

Responsibilities:

- Event setup
- Participant management
- Category configuration
- Checkpoint configuration
- Result publication

## Live Leaderboard

Responsibilities:

- Display ranking
- Display checkpoint passage
- Display participant status
- Receive realtime updates

# 5. Data Flow

## Step 1

RFID Tag Detected

Hardware produces:

```
{
  "epc": "E20000123456789012345678",
  "rssi": -52
}
```

## Step 2

Decoder receives read.

Decoder adds:

```
{
  "timestamp": "2026-06-14T08:30:15.123Z",
  "readerId": "CP1",
  "deviceId": "IZT-UHF-001"
}
```


## Step 3

Decoder sends payload.


```
{
  "eventId": "kostrad-run-2026",
  "deviceId": "IZT-UHF-001",
  "readerId": "CP1",
  "epc": "E20000123456789012345678",
  "rssi": -52,
  "timestamp": "2026-06-14T08:30:15.123Z"
}
```


## Step 4

Timing API stores payload into:

`raw_timings`
table.

## Step 5

Result Engine processes timing.

System finds participant mapping.

```
EPC
↓
Participant
↓
Bib Number
↓
Category
```

## Step 6

Leaderboard updates in realtime.

# 6. API Specification

## Endpoint

```
POST /api/v1/timing/raw
```

Headers:

```
Authorization: Bearer <API_KEY>
Content-Type: application/json
```

# 7. Payload Specification

## Required Fields


```
{
  "eventId": "kostrad-run-2026",
  "deviceId": "IZT-UHF-001",
  "readerId": "CP1",
  "epc": "E2000012345678901234567 8",
  "timestamp": "2026-06-14T08:30:15.123Z"
}
```


## Optional Fields


```
{
  "rssi": -52,
  "antenna": 1,
  "battery": 87,
  "temperature": 32.4
}
```


# 8. Database Design

## events

| field | type |
|---|---|
| id | uuid |
| name | varchar |
| start_date | datetime |

| | |
|---|---|
| status | varchar |

# participants

| field | type |
|---|---|
| id | uuid |
| event_id | uuid |
| bib_number | varchar |
| full_name | varchar |
| category | varchar |

# participant_tags

| field | type |
|---|---|
| id | uuid |
| participant_id | uuid |
| epc | varchar |

One participant may have multiple tags.

# raw_timings

| field | type |
|---|---|
| id | bigint |
| event_id | uuid |
| device_id | varchar |
| reader_id | varchar |
| epc | varchar |

| | |
|---|---|
| timestamp | datetime |
| rssi | int |

Raw timing records are immutable.

## processed_timings

| field | type |
|---|---|
| id | bigint |
| participant_id | uuid |
| checkpoint_id | uuid |
| timing | datetime |
| lap | int |

Generated by Result Engine.

# 9. Duplicate Detection

Duplicate detection must occur before result calculation.

Default rule:

Same:

- Event
- EPC
- Reader

within:

```
3 seconds
```

Store:

First Read Only.

Configuration must be adjustable.

# 10. Result Calculation Engine

Responsibilities:

- EPC Matching
- Checkpoint Validation
- Lap Detection
- Ranking Calculation
- Category Ranking
- Overall Ranking
- Net Time
- Gun Time

Result Engine must support recalculation.

# 11. Offline Mode

Decoder must support:

## Local Queue

Store unsent timing records.

## Retry Logic

Retry every:

```
5 seconds
```

until successful.

## Guaranteed Delivery

Timing records must not be discarded.

# 12. Real-Time Communication

Recommended:

SignalR

Alternative:

WebSocket

Events:

```
TimingReceived
ParticipantPassed
```

```
LeaderboardUpdated
RaceFinished
```

# 13. Security

Each event has:

```
Event ID
API Key
```

Authentication required for all timing submissions.

Rate limiting should be enabled.

All communication must use HTTPS.

# 14. Scalability Target

Minimum target:

Participants:
10,000+

Concurrent viewers:
5,000+

Timing reads:
500 reads/sec

Leaderboard latency:
< 1 second

API response:
< 200 ms

# 15. Future Roadmap

Planned support:

- Marathon
- Road Run
- Trail Run
- Cycling
- Duathlon
- Triathlon
- OCR

- Corporate Challenge

Future integrations:

- Strava
- Garmin
- Coros
- Apple Health
- Google Fit

End of Document
IZT Race Technology Architecture Specification v1.0